Product Requirement Document (PRD)

Project: Bruno — AI Gym Buddy



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1. Executive Summary & Value Proposition

Most fitness apps fail because they turn into tedious spreadsheets demanding dozens of manual inputs mid-session, or generic chatbots that do not understand real biomechanics.

Bruno is an autonomous, mascot-driven workout programming app. Framed around Bruno—a wise, chalked-up heavyweight bear carrying an old-school lifting notebook—the app's primary mission is to engineer, adapt, and evolve customized lifting programs tailored to a user's available gym hardware, schedule constraints, and recovery capacity. Bruno replaces distracting real-time tracking with high-leverage exercise programming, smart machine swaps, and automated weekly progressive overload.

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2. Target Audience & Core Persona

- **Core User**: Intermediate lifters, busy professionals, and regular gym-goers who want structured progressive overload without having to design their own programs or log every set mid-workout.
- **Core Value**: "Tell me exactly what to lift today based on my gear, progress my weights when I get stronger, and don't make me stare at my phone between sets."
- **Mascot Personality**: Calm authority, grandfatherly strength wisdom, zero fluff. Bruno explains the why behind program choices and calibrations.

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3. Product Scope & Features

Feature 1: Gym Environment & Physical Profile

- **Gym Hardware Profiles**:

    - Users can save multiple training environments (e.g., Commercial Gym, Apartment Gym, Home Dumbbells).
    - Equipment checklist: Max dumbbell weight available, barbell and squat rack access, cable stack and attachments, Smith machine, pull-up bar.
- **Physical Constraints & Injury Blacklist**:
    - Users can flag movement restrictions or joint pain (e.g., lower back sensitivity, shoulder impingement).
    - The app automatically filters out contraindicated exercises and selects biomechanically sound alternatives (e.g., swapping Barbell Back Squats for Belt Squats or Bulgarian Split Squats).
- **Schedule & Time Budget**:
    - Training frequency options (3, 4, 5, or 6 days per week).
    - Session duration limits (30 min, 45 min, 60 min) to ensure workouts fit real daily schedules.

Feature 2: Personalized Workout Program Generation

- **Periodized Routine Generation**:
    - Builds complete 4- to 6-week structured training blocks matching target splits (Push/Pull/Legs, Upper/Lower, Full Body).
    - Delivers exact targets for every exercise:
        - Prescribed working sets and rep targets (e.g., 3 sets of 8-10 reps).
        - Target starting weight (kg) based on user strength baseline.
        - Target effort level (RPE / Reps in Reserve).
- **Weekly Volume Balance**:
    - Ensures optimal weekly direct set volume (10–16 working sets per major muscle group) to maximize progress without junk volume.
- **1-Tap "Machine Busy" Swap**:
    - If a machine or bench is occupied at the gym, tapping the exercise surfaces 3 immediate replacements that match:
        1. The exact same primary muscle group.
        2. The same movement pattern (e.g., Seated Cable Row to Chest-Supported Dumbbell Row).
        3. The equipment currently available in the active gym profile.

Feature 3: 30-Second Post-Workout Logging

- **Frictionless Completion Log**:
  - No active timers, stopwatches, or mid-workout set checkers.
  - After the session, the user logs top weights/reps achieved or taps "Hit Prescribed Targets" to complete the session in one click.
- **Fatigue & Recovery Feedback**:
  - Quick post-session sliders for overall effort rating (RPE 1–10) and joint soreness level.

Feature 4: Progressive Overload & Calibration Engine

- **Automated Progression Logic**:
  - When a user hits the target rep range across all working sets, Bruno automatically increases the target weight by +2.5 kg (upper body) or +5 kg (lower body) for the following week.
  - If a user misses target reps, Bruno keeps the weight steady and adjusts working set volume.
- **Auto-Deload Detection**:
  - Automatically schedules a 1-week recovery deload (reduced weight and volume) after 4–6 weeks of heavy training or when cumulative joint fatigue is flagged.

Feature 5: Mascot Coaching & Feedback

- **Pre-Session Briefing**:
  - A concise note on the home screen setting the goal for today's workout: "Last week you hit 22.5 kg on Incline Dumbbell Press for 10 reps easily. Today Bruno bumped you to 25 kg for 8. Control the descent."
- **Weekly Review & Verdict**:
  - An end-of-week summary card showing total weight moved, volume breakdown by muscle group, and Bruno's feedback on consistency.
- **Mascot Dashboard Presence**:

    - Bruno appears on the dashboard with visual states reflecting adherence (e.g., Studying Journal, Approving Nod, Waiting with Watch).

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4. Key Success Metrics

- **Generation Speed**: Complete 4-week program generated in under 3 seconds after profile setup.
- **Hardware Accuracy**: 100% of prescribed exercises match the user's selected gym equipment.
- **Session Logging Speed**: Workouts can be logged and saved in under 30 seconds.
- **Progression Accuracy**: Automatic weight increases applied correctly to the following week whenever target reps are hit.